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Geodesic Distance Literature Map

Date: 2026-05-04

This memo maps the broader, mostly density-unaware literature on estimating intrinsic or geodesic distance from finite data. It is intentionally separate from the density-aware lane. Density enters here mainly through sampling, neighbor radii, local scales, and graph connectivity; it is not the primary edge-cost target unless stated explicitly.

The central lesson for gflow is simple but important: graph topology and edge weights are different design decisions. A shortest-path distance can only be as good as the graph on which paths are allowed to travel. Reweighting a bad shortcut may help if the weight detects that it is unsupported, but a cleaner workflow first asks whether the neighbor graph should have admitted the edge at all.

Executive Map

The classical graph-geodesic estimator is the Isomap distance:

$$d_G(i, j) = \min_{\pi: i \rightarrow j} \sum_{(u, v) \in \pi} \|x_u - x_v\|.$$

The graph G is usually a k -nearest-neighbor graph or an epsilon/radius graph, and the edge weights are local ambient distances. If the data densely sample a smooth manifold, and the neighborhood scale shrinks slowly enough, then these graph shortest paths can

converge to the manifold geodesic distance. This is the borrowed backbone behind many practical data-geodesic ideas, starting with Isomap ([Tenenbaum, de Silva, and Langford 2000](#)) and its theoretical analysis ([Bernstein et al. 2000/2001](#)).

For gflow, the competing families should be evaluated fairly:

- ordinary directed/symmetrized kNN shortest paths;
- mutual kNN and shared-neighbor graphs;
- adaptive-k, local-scale, radius, and hybrid graphs;
- pruning by geometric empty-region rules, local alternative paths, or topology;
- connectivity repair by MST, component bridges, landmarks, or scaffold edges;
- diffusion, resistance, PHATE, UMAP, and pseudotime distances as comparators, but not automatically as geodesic distances.

No method family is the answer by definition. CMST/MST completion is best viewed as one connected scaffold strategy. It can be useful when finite graph distances, edge attribution, and bridge diagnostics matter, but it can also impose tree artifacts or false bridges. A well-tuned kNN or mKNN graph may be the cleaner geodesic estimator in some regimes.

Basic Objects and Vocabulary

Let

$$X_n = \{x_1, \dots, x_n\} \subset \mathbb{R}^D$$

be sampled near a lower-dimensional object M , often imagined as a compact d -dimensional Riemannian manifold, stratified manifold, tube, curve network, or branching biological state space.

A finite weighted graph is

$$G = (V, E, w), \quad V = \{1, \dots, n\}, \quad w_{ij} \geq 0.$$

The graph shortest-path metric is

$$d_G(i, j) = \min_{i=v_0, \dots, v_m=j} \sum_{r=0}^{m-1} w_{v_r v_{r+1}},$$

with $d_G(i, j) = \text{infinity}$ if i and j are disconnected. On each connected component with nonnegative weights and no zero-weight identifications, this is a metric. With zero weights or directed/asymmetric reachability, it may become a pseudometric or a directed quasi-distance.

The continuous target, when it exists, is the Riemannian geodesic distance

$$d_M(x, y) = \inf_{\gamma: x \rightarrow y} \int_0^1 \sqrt{g_{\gamma(t)}(\gamma'(t), \gamma'(t))} dt.$$

The data problem is not just “estimate this equation.” Biological data add finite-sample gaps, high-dimensional nuisance variation, batch effects, count noise, compositional transforms, branches, loops, boundaries, outliers, and scientific ambiguity about whether a missing region is an unobserved transition or a true forbidden state.

Historical Spine

1. **Computational geometry and graph shortest paths.** Dijkstra-style graph shortest paths are older than manifold learning, but proximity graphs such as Delaunay triangulations, Gabriel graphs, relative neighborhood graphs, beta skeletons, and geometric spanners supplied a vocabulary for sparse graphs that preserve local geometry or Euclidean stretch.
2. **Isomap made graph geodesics central to manifold learning.** Isomap builds a kNN or epsilon graph, weights edges by local Euclidean distances, computes all-pairs shortest paths, and then uses classical MDS on those approximate geodesic distances ([Science 2000](#)).
3. **Theory clarified the asymptotic promise.** Bernstein, de Silva, Langford, and Tenenbaum analyzed when graph distances approximate manifold geodesics under dense enough sampling, bounded curvature, and appropriate neighborhood scale ([graph approximation report](#)). Later random geometric graph theory made connectivity and shortest-path behavior more precise ([Penrose 2003](#)).
4. **Failure modes were recognized almost immediately.** Too small a neighborhood disconnects the graph or produces tree-like detours. Too large a neighborhood creates “short circuits” across folds, holes, branches, gaps, or noisy off-manifold points. Balasubramanian and Schwartz highlighted topological instability in early Isomap discussions; modern software still exposes the same parameter tradeoff.
5. **Scalable variants replaced all-pairs geodesics with landmarks.** Landmark Isomap / Landmark MDS embeds a subset of landmarks and places the remaining points from landmark distances, reducing memory and time at the cost of landmark-selection sensitivity ([de Silva and Tenenbaum 2004](#)).
6. **Out-of-sample extension became a separate issue.** A new point must be attached to the training graph, distances from it to training points must be estimated, and the embedding or downstream distance must be extended without rebuilding everything. Bengio et al. framed Isomap, MDS, LLE, eigenmaps, and spectral clustering as eigenfunction extensions ([NeurIPS 2003](#)).
7. **Applied biology mostly adopted graphs, not pure Isomap distances.** Single-cell workflows widely use kNN graphs, MSTs, principal graphs, diffusion maps, PAGA, and pseudotime graph distances. Microbiome and metagenomics workflows more often use beta-diversity, Aitchison/UniFrac, ordination, PHATE, UMAP, and network/TDA methods than explicit kNN shortest-path distance matrices.

Construct 1: Classical Isomap and kNN/Epsilon Graph Geodesics

Motivation and Problem Setting

Isomap was introduced for nonlinear dimensionality reduction when pairwise ambient distances are misleading because the data lie on or near a curved low-dimensional manifold. The problem setting is geometric rather than probabilistic: preserve intrinsic

distances on the manifold, then find a low-dimensional Euclidean representation that best matches those distances.

Definition

Choose a neighborhood graph:

$$E_k^{\text{dir}} = \{(i, j) : x_j \in N_k(x_i)\},$$

or

$$E_\varepsilon = \{(i, j) : \|x_i - x_j\| \leq \varepsilon\}.$$

For undirected Isomap, directed kNN is usually symmetrized by union:

$$E_k^\cup = \{(i, j) : j \in N_k(i) \text{ or } i \in N_k(j)\}.$$

Weights are usually

$$w_{ij} = \|x_i - x_j\|.$$

The graph-geodesic estimate is

$$\hat{d}_M(x_i, x_j) = d_G(i, j).$$

Isomap then embeds with classical scaling:

$$B = -\frac{1}{2}HD_G^{\circ 2}H, \quad H = I - \frac{1}{n}\mathbf{1}\mathbf{1}^T,$$

and takes the leading eigenvectors of B.

Metric type: weighted graph metric on connected components; continuous geodesic estimator under assumptions; embedding distance only after MDS.

Parameters and Practical Defaults

The main parameter is neighborhood scale. `k` controls degree; `epsilon` controls a global radius. There is no universal default. Current scikit-learn Isomap exposes `n_neighbors=5`, `radius=None`, `path_method='auto'`, and `n_components=2` by default ([scikit-learn Isomap docs](#)). Single-cell software often uses larger graph neighborhoods for other purposes: Scanpy defaults to `n_neighbors=15` for `scanpy.pp.neighbors` ([Scanpy docs](#)), while Seurat `FindNeighbors` defaults to `k.param=20` and often computes a shared-nearest-neighbor graph ([Seurat docs](#)). Those are adoption signals, not geodesic-optimality guarantees.

Common selection heuristics:

- sweep `k` or `epsilon`;
- require one large component but inspect bridges;
- use residual variance or embedding stress for Isomap;
- check neighborhood overlap against known latent coordinates in simulations;
- check false-shortcut edges across folds, holes, branches, or gaps;
- prefer stability across a range rather than a single lucky parameter.

Asymptotic and Finite-Data Behavior

For a smooth compact manifold with enough samples and an appropriate shrinking radius, local Euclidean edge lengths approximate local manifold arc lengths, and shortest paths on the graph approximate continuous geodesics. The intuition is: each true geodesic can be covered by small sample-to-sample steps, and each graph edge is short enough that its ambient chord is close to its intrinsic local distance. The theory requires sampling density bounded away from zero on the support, a scale larger than the covering radius, and local geometry controlled by curvature/reach.

Finite data violate these assumptions in exactly the ways `gflow` cares about: uneven sampling, high-dimensional noise, off-manifold thickness, true gaps, branch points, and boundaries.

Strengths

- Directly estimates an interpretable graph geodesic.
- Simple to implement and inspect.
- Edge/path attribution is straightforward.
- Works well on clean S-curves, spirals, circles, and tubes when k is tuned.
- Separates local metric choice from downstream regression or embedding.

Failure Modes

- **Shortcuts:** an edge across a fold, crescent, annulus, branch, gap, or figure-eight self-approach can shorten many paths.
- **Disconnectedness:** small k or small `epsilon` fragments the graph, yielding infinite distances.
- **Tree artifacts:** too sparse a graph approximates a traversal tree rather than the manifold.
- **Boundary effects:** boundary points have fewer neighbors, can be pulled into long detours, or can be connected across periodic boundaries incorrectly.
- **Holes and voids:** a missing sector can be treated as either a real hole or a sample gap; the graph does not know which prior is intended.
- **Noise:** off-manifold radial or nuisance variation can create ambient chords that are not intrinsic adjacencies.
- **Nonuniform sampling:** fixed k uses larger radii in sparse regions and smaller radii in dense regions; fixed radius does the reverse with degree.

`gflow` Diagnostics

For every graph-geodesic estimator, `gflow` should report:

- component count and largest component fraction;
- edge-length distribution by graph family;
- degree distribution and high-degree hubs;
- cycle rank and whether a circle-like structure is path-like or loop-like;
- false-shortcut rate on synthetic truth;
- long bridge edges and their path usage;
- sensitivity over k , `epsilon`, preprocessing, and PCA dimension;
- shortest-path attribution for influential downstream regression pairs.

Relevance: the classical estimator is the baseline to beat. On noisy circles, tubes, gaps, branches, and compositional gradients, it makes the right question visible: which topology plus which weights recover the latent distances useful for gflow regression and gradient-flow summaries?

Construct 2: Landmark Isomap, Scalable Isomap, and Out-of-Sample Extension

Landmark Isomap

Landmark Isomap chooses a subset L of $m \ll n$ landmark points, computes shortest-path distances from landmarks to all samples, embeds the landmarks by MDS, and places non-landmarks from their landmark-distance profiles. Landmark MDS uses the fact that distances to enough well-spread landmarks can locate a point in the embedding.

Let D_{LL} be landmark-to-landmark graph distances and D_{LX} be landmark-to-all graph distances. Classical MDS is applied to D_{LL} ; nonlandmark points are projected by a Nyström-like / triangulation formula from D_{LX} .

Metric type: the underlying distances are graph geodesics; the scalable output is an approximate embedding and approximate coordinate extension.

Strengths:

- avoids storing or eigendecomposing the full $n \times n$ matrix;
- can support large biological data sets;
- gives natural landmark/witness diagnostics.

Failure modes:

- poor landmark coverage misses branches, rare states, holes, or sparse transitions;
- shortest-path errors from a bad graph are inherited;
- landmark distances can smooth over local defects.

gflow relevance: landmarks are useful for large n , but rare biological states and compositional tails should be actively sampled as landmarks rather than left to random chance.

Out-of-Sample Extension

For a new sample x_* , one common Isomap extension attaches it to the training graph through its nearest training neighbors:

$$\hat{d}_G(*, j) = \min_{i \in N_k(*)} \|x_* - x_i\| + d_G(i, j).$$

The new squared-distance vector is then centered and projected into the learned MDS/kernel-PCA coordinates. This is the practical logic exposed in scikit-learn's Isomap transform, which links query points to the training geodesic graph and projects the resulting kernel onto the training embedding ([scikit-learn Isomap docs](#)). Bengio et al. give the broader eigenfunction-extension framing ([NeurIPS 2003](#)).

For gflow, out-of-sample distance extension matters more than embedding extension. New samples should report which training edges or bridge components control their geodesic attachment; otherwise, a single off-manifold query can silently inherit a misleading downstream gradient.

Construct 3: kNN Graph Variants and Local Topology Choices

The neighborhood graph is the main finite-data object. Edge weights can be changed later, but an absent edge cannot be used and an admitted shortcut can dominate all-pairs paths.

Directed kNN

Definition:

$$(i, j) \in E_k^{\text{dir}} \iff x_j \in N_k(x_i).$$

Metric type: directed graph distance if paths respect edge direction. This is not symmetric:

$$d_G(i, j) \neq d_G(j, i)$$

in general.

Use case: directed kNN is natural when modeling flow, velocity, or asymmetric reachability. For ordinary geodesic distance, it is usually an intermediate construction before symmetrization.

Failure mode: directed reachability can create one-way finite distances and break ordinary metric assumptions in regression unless explicitly intended.

Symmetrized kNN

Union symmetrization:

$$\{i, j\} \in E_k^{\cup} \iff j \in N_k(i) \text{ or } i \in N_k(j).$$

This keeps edges if either endpoint selected the other. It is more connected and can preserve sparse-region reachability, but admits asymmetric hub edges.

Intersection / mutual kNN:

$$\{i, j\} \in E_k^{\cap} \iff j \in N_k(i) \text{ and } i \in N_k(j).$$

Mutual kNN is conservative. It often reduces hubness and false long-radius edges from sparse points into dense regions, but it fragments easily.

Metric type: graph construction; becomes a graph metric after weights and shortest paths are defined.

gflow relevance: this is a direct candidate family. The noisy-circle note already observed a plausible division: iKNN/symmetrized kNN can recover loops but may become dense; mKNN can be geometrically cleaner but disconnected.

Shared Nearest Neighbor Graphs

Shared-neighbor methods weight an edge by overlap between neighbor sets:

$$s_{ij} = \frac{|N_k(i) \cap N_k(j)|}{|N_k(i) \cup N_k(j)|}$$

or related counts. Seurat's SNN graph uses neighborhood overlap and prunes low Jaccard-weight edges by default ([Seurat docs](#)).

Metric type: similarity graph; not a geodesic distance unless transformed into edge lengths and used in shortest paths.

Strength: robust to local high-dimensional noise and hubness.

Failure: shared-neighbor weights can emphasize cluster membership over intrinsic arclength, which may be good for clustering and bad for distance estimation.

Radius / Epsilon Graphs

Definition:

$$\{i, j\} \in E_\epsilon \iff \|x_i - x_j\| \leq \epsilon.$$

Metric type: graph construction; weighted shortest paths estimate geodesics under classical random geometric graph assumptions.

Strength: simple and theoretically clean for bounded-density samples.

Failure: a global radius gives high degree in dense regions and isolated points in sparse regions. In biological data with uneven sampling, this can be worse than fixed k.

Hybrid kNN/Radius Graphs

Hybrid admission rules combine local degree and maximum edge length, for example:

$$\{i, j\} \in E \iff (j \in N_k(i) \text{ or } i \in N_k(j)) \text{ and } \|x_i - x_j\| \leq r_{\max}.$$

or

$$\|x_i - x_j\| \leq c \max(r_i, r_j), \quad r_i = \|x_i - x_{i(k)}\|.$$

Metric type: graph construction. The second rule is local-scale or adaptive radius rather than global radius.

Strength: prevents some long sparse-region kNN shortcuts while preserving minimum degree.

Failure: local scale must be capped or robustified around outliers; otherwise outlier neighborhoods can authorize exactly the bridges one wants to flag.

Locally Scaled Graphs

Self-tuning spectral clustering defines a local scale σ_i , often the distance to the k th neighbor, and uses affinities like

$$A_{ij} = \exp \left(-\frac{\|x_i - x_j\|^2}{\sigma_i \sigma_j} \right).$$

The original motivation was multi-scale clustering and clutter robustness ([Zelnik-Manor and Perona 2004](#)).

Metric type: kernel/spectral graph. It is not a shortest-path geodesic unless one deliberately converts affinities into path costs, for example $w_{ij} = -\log A_{ij}$ or a locally scaled edge length.

gflow relevance: local scales are natural diagnostics and edge-admission tools. They should be kept conceptually separate from density-aware ℓ/ρ^α edge costs, because local scaling can either exploit density, normalize it away, or merely adapt the graph to uneven sampling.

Adaptive Neighborhoods

Adaptive-neighborhood methods try to infer the right local graph degree from the data rather than fixing k . IAN starts from a conservative Gabriel graph, then iteratively sparsifies/adapts neighborhoods using a global optimization and outlier checks, with applications to nonlinear dimension reduction, geodesic computation, and dimension estimation ([Dyballa and Zucker 2023](#), [arXiv](#)).

Metric type: graph construction; final graph distances may be geodesic estimators.

gflow relevance: adaptive k is likely important for uneven noisy circles, sparse transitions, branches, and compositional data where fixed-neighborhood rules confound sampling density with biology.

Construct 4: Connectivity Repair and MST Scaffolds

MST and CMST as Scaffolds

The Euclidean MST on X_n is

$$T = \arg \min_{\text{spanning trees } T} \sum_{\{i,j\} \in T} \|x_i - x_j\|.$$

It guarantees connectedness with $n-1$ edges. CMST-style completion starts from an MST and adds local edges under a threshold or rule:

$$E_{\text{CMST}} = E_{\text{MST}} \cup E_{\text{local}}.$$

Metric type: connected graph shortest-path metric after weights are assigned.

Strengths:

- finite distances for all pairs;
- long forced bridges are inspectable;
- sparse and interpretable;
- useful when downstream methods require one connected graph.

Failures:

- an MST is not a manifold skeleton in general;
- it can break loops into paths;
- it can force false bridges between true components;
- adding MST edges to an already good kNN/mKNN graph can shorten paths through tree artifacts.

gflow position: MST/CMST is a candidate scaffold and diagnostic, not a privileged source of intrinsic geometry.

Component Repair for Conservative Graphs

A conservative graph such as mKNN may fragment. Instead of unioning with the full MST, one can connect components by selected bridges:

$$e_{ab}^* = \arg \min_{i \in C_a, j \in C_b} \|x_i - x_j\|,$$

possibly subject to local-scale or geometric plausibility constraints:

$$\|x_i - x_j\| \leq c \max(r_i, r_j), \quad \text{or} \quad \frac{\|x_i - x_j\|}{\text{median local edge}} \leq c.$$

Repair can also use a component-level MST, where vertices are components and candidate edges are nearest inter-component bridges.

Metric type: graph repair before shortest paths.

Strength: preserves the low-shortcut behavior of mKNN while making distances finite.

Failure: if the component split represents a true biological separation, repairing it erases meaningful disconnectedness.

Diagnostics: number of repaired components, bridge length quantiles, bridge path usage, downstream sensitivity to removing or inflating bridge weights, and whether bridges cross known latent gaps in simulations.

Construct 5: Pruning and Shortcut Control

Pruning asks whether an admitted edge should remain. This is separate from edge reweighting.

Local Alternative-Path Pruning

An edge $e=\{i, j\}$ can be removed if the graph without it still has a short alternative path:

$$d_{G \setminus e}(i, j) \leq \tau w_{ij}.$$

For τ near 1, this removes redundant edges while preserving local path lengths. A greedy spanner uses the dual rule: add an edge only if no existing path is short enough.

Metric type: topology modification before graph shortest paths.

Strength: reduces dense kNN/iKNN shortcuts and can preserve cycle geometry with fewer edges.

Failure: if the alternate path itself contains a shortcut, the rule ratifies bad topology. If τ is too strict, it leaves a near-tree; if too loose, it may overprune legitimate local supports.

gflow diagnostics: removed-edge attribution, stretch distribution, component check after each pruning level, and false-shortcut rate on known latent cases.

Relative Neighborhood Graph

The relative neighborhood graph connects i and j when there is no third point closer to both endpoints than they are to each other:

$$\{i, j\} \in E_{\text{RNG}} \iff \nexists k : \max\{\|x_i - x_k\|, \|x_j - x_k\|\} < \|x_i - x_j\|.$$

Toussaint introduced RNGs as a proximity structure for finite planar sets ([Toussaint 1980 record](#)). RNGs are subgraphs of Delaunay triangulations in the plane and contain the Euclidean MST, hence are connected for planar Euclidean point sets in general position ([overview](#)).

Metric type: sparse proximity graph; shortest paths on it are graph geodesics but not automatically manifold-geodesic estimators.

Strength: removes many edges with obvious local alternatives.

Failure: can be too sparse and has no direct adaptation to high-dimensional noise, compositional metrics, or manifold folds.

Gabriel Graph

The Gabriel graph connects i and j if the closed ball with diameter $x_i - x_j$ contains no other sample point:

$$\{i, j\} \in E_{\text{Gabriel}} \iff \nexists k : \left\| x_k - \frac{x_i + x_j}{2} \right\| < \frac{1}{2} \|x_i - x_j\|.$$

Gabriel and Sokal introduced the graph in geographic variation analysis; it is also a beta skeleton and a subgraph of the Delaunay triangulation in the plane ([overview](#)).

Metric type: empty-ball proximity graph.

Strength: scale-free local criterion; useful conservative starting graph.

Failure: empty-ball tests in high dimensions are fragile because points are sparse and distances concentrate; a Gabriel edge can still be a manifold shortcut across a fold if the ambient ball is empty.

Delaunay-Based Graphs

Delaunay triangulation connects points whose Voronoi cells touch. In low dimensions it is a rich proximity supergraph. Many proximity graphs, including Gabriel and RNG, are subgraphs of Delaunay in the plane.

Metric type: proximity graph; weighted shortest paths can be used as graph geodesics.

Strength: good computational geometry foundation and spanner properties in Euclidean planar settings.

Failure: high-dimensional Delaunay is often infeasible and may be too dense; ambient Delaunay does not know the manifold topology.

Beta Skeletons

Beta skeletons use an empty lens or circle region controlled by beta; Gabriel is the beta=1 case and the relative neighborhood graph is a common beta=2 lune-based case. Increasing beta usually makes the graph more conservative ([beta skeleton overview](#)).

Metric type: geometric pruning family.

Strength: tunable shortcut control.

Failure: planar Euclidean intuition may not transfer to high-dimensional biological feature spaces without preprocessing.

Geometric Spanners

A weighted graph is a t -spanner of a metric space if

$$d_G(i, j) \leq t d(i, j)$$

for all pairs. Greedy spanners add an edge only when the current graph lacks a path of length at most t times the direct distance. Theta/Yao graphs use cones around each point to keep a bounded number of directional neighbors. Geometric spanners were developed for sparse networks that preserve Euclidean distances ([geometric spanner overview](#)).

Metric type: sparse graph with bounded stretch relative to a chosen base metric, usually Euclidean.

Important caution: Euclidean spanner guarantees are not manifold-geodesic guarantees. A graph that well preserves ambient Euclidean distances may be exactly the graph that preserves harmful chords.

gflow relevance: spanner logic is valuable when applied to a local graph or latent/oracle metric, not blindly to ambient Euclidean all-pairs distances.

Witness and Landmark Graphs

Witness graphs and witness complexes use a smaller landmark set L and allow nonlandmark points to witness relationships among landmarks. A simple graph version connects landmarks a, b when some witness x has both among its near landmarks:

$$\{a, b\} \in E \iff \exists x : a, b \in N_m^L(x).$$

Metric type: sparse topology surrogate; graph geodesic only after edge weights are defined.

Strength: scalable and useful for rare-state coverage if landmark selection is designed.

Failure: compresses cell/sample-level paths; branch tips and small holes can be lost if no landmark/witness supports them.

Topology-Aware Pruning

Topology-aware pruning uses homology, cycle persistence, branch consistency, or cluster abstraction to decide whether edges preserve or destroy intended structure. In single-cell analysis, PAGA estimates connectivity of manifold partitions from a symmetrized kNN-like graph and provides a topology-preserving abstracted graph ([PAGA Genome Biology 2019](#)).

Metric type: usually graph abstraction or topology diagnostic; not necessarily a sample-level geodesic metric.

gflow relevance: topology-aware pruning could flag false circle chords, figure-eight crossovers, branch-to-branch shortcuts, and gap bridges before distance estimation.

Construct 6: Theoretical Foundations

Convergence of Graph Shortest Paths

The classical convergence story requires:

- samples lie on or close to a compact smooth manifold;
- sampling density is bounded above and away from zero on the support;
- the neighborhood radius ϵ_n tends to zero but remains above the sample covering radius;
- local chords approximate local manifold distances;
- curvature, reach, and boundary do not destroy local Euclidean approximations.

In this regime, graph shortest paths over ϵ_n or local graphs converge to manifold geodesics. Bernstein et al. provided an early proof for Isomap-style claims. Later work generalized shortest-path approximation on sampled subsets, including curvature-constrained variants ([Arias-Castro and Le Gouic 2017](#)).

Random Geometric Graph Connectivity

A random geometric graph connects points within radius r_n . Connectivity thresholds scale like

$$r_n \asymp \left(\frac{\log n}{n} \right)^{1/d}$$

up to constants and boundary/density effects. Penrose's monograph is the standard reference for rigorous random geometric graph theory ([Penrose 2003](#)).

For kNN graphs, the right order for connectivity is also logarithmic in n in classical uniform settings. Balister, Bollobas, Sarkar, Walters, and related work sharpen constants in planar random kNN graphs; a recent survey-style introduction notes that $k \sim \log n$ is the right order for connectivity ([PMC review/introduction](#)).

Practical implication: fixed small k is not an asymptotic connectivity rule. It is a finite-data modeling choice. In single-cell workflows, fixed defaults like 15 or 20 are pragmatic, not theoretical guarantees.

Density and Edge Weight Regimes

Alamgir and von Luxburg analyzed shortest-path limits in random kNN graphs and showed that unweighted kNN shortest paths can induce an undesirable density-dependent geometry, preferring low-density regions in a way that is opposite to many machine-learning intuitions ([ICML 2012 PDF](#), [arXiv](#)).

This is directly relevant to `gflow`: hop count is not a neutral geodesic distance. Euclidean edge weights are usually safer as a default for geometric distance, while density-aware or power-weighted alternatives should be named as separate metric choices.

Power-weighted shortest paths

$$d_p(i, j) = \min_{\pi: i \rightarrow j} \sum_{(u, v) \in \pi} \|x_u - x_v\|^p$$

have a density-weighted continuum limit for $p > 1$ and are covered in the density-aware report. Here they matter mainly as a warning: changing edge weights changes the target metric, not just numerical stability.

Intrinsic Dimension, Reach, Curvature, Boundary

Intrinsic dimension controls covering rates and connectivity thresholds. Reach controls how far one can move before local Euclidean neighborhoods become ambiguous across folds or self-approaches. Curvature controls chord-vs-arc error:

$$\text{arc}(i, j) - \|x_i - x_j\| = O(\kappa^2 \|x_i - x_j\|^3)$$

for small edges on a smooth curve with curvature κ . Boundaries reduce neighbor support and make symmetric local neighborhoods false. Branch points and stratified intersections violate the smooth-manifold assumptions entirely.

Practical implication: a method can be asymptotically consistent on smooth manifolds and still fail on the exact biological structures `gflow` wants: branches, basins, loops, gaps, tubes, and compositional sparse count clouds.

kNN Spanner Results

Spanner results say when a sparse graph approximately preserves shortest-path distances of another graph or metric. Little, McKenzie, and Murphy showed that kNN graphs can approximate complete-graph power-weighted shortest-path distances under suitable sampling and k scaling ([SIAM MDS 2022](#)).

For density-unaware $p=1$ Euclidean-weighted graph geodesics, spanner intuition supports using sparse local graphs for computation, but only after the target metric and admissible neighborhood topology are specified.

Practical Adoption Review

Bioinformatics and Computational Biology

General bioinformatics has used Isomap and manifold learning for visualization, feature extraction, image and microarray analysis, and disease prediction, but explicit all-pairs graph-geodesic distances are less commonly treated as the final scientific object. For example, microarray feature-selection work describes Isomap's kNN graph and Dijkstra/Floyd-Warshall geodesic matrix as part of a manifold-learning pipeline ([PMC example](#)). Metagenomic deep-learning work has used Isomap as a 2D representation method, again defining geodesic distances by shortest paths on a kNN graph ([Cell Reports Methods 2022](#)).

Transferable lesson: practical users often consume the embedding, not the distance matrix. `gflow` is more distance-forward, so it needs stronger diagnostics than typical visualization papers provide.

Microbiome, Metagenomics, and Compositional Count Data

The dominant microbiome distance ecosystem is not Isomap-style graph geodesics. It is beta-diversity and compositional geometry: Bray-Curtis, Jaccard, Jensen-Shannon, weighted/unweighted UniFrac, generalized UniFrac, robust Aitchison, DEICODE, ordination, and PERMANOVA-style workflows.

Evidence for routine kNN shortest-path geodesics in microbiome analysis is thin. Isomap appears as a representation option in some metagenomic machine learning workflows, and PHATE/tmap/diffusion/TDA approaches show that graph geometry is acceptable in the domain, but explicit geodesic distance matrices are not standard microbiome outputs.

`gflow` implications:

- use Aitchison/CLR/robust Aitchison or other compositional preprocessing before Euclidean neighbor search;
- compare to UniFrac and beta-diversity baselines rather than only Euclidean kNN;
- treat zeros, rare taxa, library size, and batch effects as topology risks;

- inspect whether graph paths follow ecological gradients or merely sequencing depth and compositional sparsity.

Single-Cell Genomics and Trajectory Inference

Single-cell analysis is the strongest adoption area for graph paths, but the term “geodesic” often refers to distance along an inferred trajectory or principal graph rather than all-pairs Isomap distance on the original kNN graph.

Examples:

- Monocle 2 learns a principal tree and calculates pseudotime as geodesic distance from a root after projecting cells to the principal graph ([Monocle 2 / DDRTree PMC](#)).
- Reviews of trajectory methods summarize many MST, principal-graph, and kNN graph workflows where pseudotime is a distance along a learned graph or curve ([scTEP BMC Bioinformatics 2023](#), [Slingshot comparison table](#)).
- Wanderlust uses ensembles of kNN graphs and shortest-path-like temporal distances; Wishbone and DPT use diffusion geometry; PAGA abstracts a kNN-like graph into a topology-preserving partition graph; Palantir uses diffusion maps and shortest paths in diffusion-component space in parts of its workflow ([VIA review/background](#)).

This is a strong adoption signal for graph-based geometry but a caution for terminology. A root-to-cell pseudotime on a principal tree is a graph geodesic on that tree; it is not the same as all-pairs manifold geodesic distance on the sample cloud.

gflow implications:

- principal-graph distances should be comparators for branches and basins;
- all-pairs graph geodesics may capture more geometry than a fitted tree, but can also overfit noise;
- pseudotime methods provide practical diagnostics: root sensitivity, branch assignment, projection residuals, and lineage stability.

Spatial Transcriptomics

Spatial transcriptomics often uses graph distances in physical space or segmentation graphs. A recent point-cloud segmentation framework computes geodesic distance as graph shortest path between RNA nodes and cell centroids using Dijkstra’s algorithm ([PMC spatial transcriptomics example](#)).

For cell-state manifolds in expression space, adoption appears more diffusion, neighbor graph, and graph neural network oriented than explicit Isomap-style geodesic-distance oriented.

gflow implications: distinguish physical-space geodesics, expression-space geodesics, and tissue-neighborhood graph distances. They answer different questions.

Ecology and Community Composition

Ecology uses distance along environmental gradients, resistance surfaces, and ordination heavily. Graph shortest paths are natural in landscape connectivity, but the graph usually represents geography, dispersal cost, or habitat resistance rather than a sample cloud in high-dimensional composition space.

Transferable lesson: when a path distance is scientifically useful, the cost surface is interpretable. For gflow microbiome gradients, the analogous cost surface might be compositional dissimilarity, ecological transition support, or sample-density-aware plausibility; it should not be left implicit.

Medical Imaging, Morphology, Shape, and Phenotyping

Shape analysis uses geodesics on meshes, heat kernels, diffusion descriptors, and intrinsic surface distances. Mesh geodesic algorithms differ from sample kNN graph geodesics because the surface connectivity is often observed or reconstructed. Graph-based discrete geodesic methods on meshes can carry accuracy guarantees when vertices are well distributed ([discrete geodesic graph example](#)).

Transferable lesson: when topology is known from a mesh, shortest paths are much safer. In gflow, topology is inferred and should be diagnosed as the main uncertain object.

Non-Biological Domains

Isomap-style geodesics are widely used in computer vision, image manifolds, robotics/path planning, remote sensing, speech/audio manifolds, and general nonlinear dimensionality reduction. Non-biological lessons transfer cleanly:

- nearest-neighbor scale selection controls success;
- holes and folds punish shortcuts;
- landmarks are necessary for scale but can miss rare modes;
- high-dimensional approximate nearest-neighbor errors can change graph topology;
- embedding quality does not prove distance quality.

Related but Distinct Distances

Diffusion Distance

Diffusion maps build a kernel graph, normalize it into a Markov transition operator P , and compare transition distributions after t steps:

$$D_t^2(i, j) = \sum_k \frac{(P^t(i, k) - P^t(j, k))^2}{\pi_k}.$$

Coifman and Lafon introduced diffusion maps as a framework relating Markov processes and geometric descriptions of data ([Coifman and Lafon 2006](#)).

Metric type: random-walk/operator distance, often represented by diffusion coordinates. It integrates over many paths; it is not a graph shortest path.

When to compare: diffusion distance is a useful robust comparator for noisy tubes and branches. It may be better for basin structure and worse for exact arc-length recovery.

Commute and Resistance Distance

For a connected weighted graph with Laplacian L , effective resistance is

$$R_{ij} = (e_i - e_j)^T L^+ (e_i - e_j).$$

Commute time is proportional to resistance:

$$C_{ij} = \text{vol}(G) R_{ij}.$$

Metric type: electrical/random-walk distance. It depends on all parallel paths, not the shortest path. In large random neighborhood graphs, commute times can be dominated by degree and volume effects rather than global geometry ([von Luxburg, Radl, and Hein 2014](#)).

gflow use: comparator and graph redundancy diagnostic, not a geodesic metric.

Heat-Kernel Distance

The manifold heat kernel satisfies Varadhan's small-time relation:

$$d_M(x, y)^2 = -\lim_{t \rightarrow 0} 4t \log H_t(x, y).$$

On graphs, heat kernels use

$$H_t = \exp(-tL).$$

Metric type: heat/operator distance or descriptor; only the small-time limit has a direct geodesic interpretation.

gflow use: multiscale stability diagnostic, especially for local neighborhoods and graph smoothness.

PHATE Potential Distance

PHATE builds an adaptive affinity graph, diffuses transition probabilities, transforms them into potentials, and embeds potential distances by MDS ([Moon et al. 2019](#), [PHATE docs](#)).

Sketch:

$$P_t = P^t, \quad D_{\text{PHATE}}(i, j) = \|U(P_t(i, \cdot)) - U(P_t(j, \cdot))\|_2.$$

Metric type: diffusion/potential distance and embedding distance. It is support-aware, but not a shortest-path geodesic.

gflow use: important comparator for biological transitions and microbiome branches, especially because it has practical adoption.

UMAP Graph Geometry

UMAP constructs a weighted kNN fuzzy simplicial set. It estimates local connectivity using `rho_i` and local scales `sigma_i`, then combines asymmetric local fuzzy graphs by fuzzy set union. The UMAP docs describe the construction as locally approximating geodesic distance at each point and combining local fuzzy simplicial sets into a global graph ([UMAP API docs](#), [UMAP overview](#)).

Metric type: fuzzy topological graph plus embedding objective. The final UMAP coordinates are embedding distances, not graph shortest-path distances. The high-dimensional fuzzy graph can be used as a weighted graph, but then one must define a path cost from membership strengths.

gflow use: comparator graph topology and local-scale diagnostic; avoid calling UMAP embedding distances geodesic distances.

Trajectory and Pseudotime Graph Distances

Pseudotime often uses distance from a root along an inferred graph, principal curve, principal tree, cluster MST, or diffusion graph. This can be a true graph geodesic on the inferred trajectory:

$$\text{ptime}(i) = d_{G_{\text{traj}}}(r, \text{proj}(i)).$$

But it is root-specific, often one-dimensional per lineage, and depends on the trajectory abstraction.

gflow use: strong downstream comparator for branches and gradients, but not a replacement for all-pairs intrinsic distance unless the intended geometry is a tree/trajectory.

Relevance to gflow Benchmarks

Noisy Circle and Thick Tube

Best stressors:

- does the graph recover a loop rather than a path?
- are radial/chord shortcuts admitted?
- does mKNN fragment at sparse angular sectors?
- does MST/CMST force a long closure bridge or add false chords?
- does pruning remove redundant chords while preserving the fundamental cycle?

Recommended candidates:

- symmetrized kNN over k sweep;
- mKNN over k sweep;
- mKNN with component repair;
- iKNN/symmetrized kNN with local alternative-path pruning;
- CMST/MST completion as connected scaffold;
- UMAP/PHATE/diffusion as comparators, not targets.

Gaps and Holes

The graph cannot decide whether a missing region is a sampling gap or a true hole. gflow should therefore report two truth interpretations when possible: continuous latent manifold with missing samples, and disconnected/forbidden support with the gap treated as real.

Diagnostics: long bridge edges, number of completion edges crossing the gap, path dependence on bridge edges, and downstream regression sensitivity when bridges are removed or inflated.

Branches and Figure-Eights

Branch points and self-approaches punish ambient-nearest-neighbor assumptions. Good diagnostics are edge branch labels, endpoint-to-endpoint distortion, false cross-branch edge rate, and whether pruning destroys or preserves true branch-point connectivity.

High-Dimensional Nuisance Variation

Neighbor search should be evaluated in raw features, PCA space, known latent space for simulations, and compositional transforms. Approximate nearest-neighbor recall is also a topology variable: a few wrong neighbors can become global shortcuts.

Compositional Data

For microbiome and metagenomics, Euclidean distances on raw relative abundance are rarely the right local metric. Candidate spaces should include CLR with pseudocount sensitivity, robust CLR/DEICODE-like representations, Aitchison distance, filtering variants, and possibly UniFrac-derived neighbor graphs.

Downstream gflow regression should track whether distance improvements in latent benchmarks translate into smoother conditional expectation estimates, more stable gradient-flow calls, and interpretable edge/path attribution.

Synthesis Table

Family	Primary role	Metric type	Should g call i geodes
Isomap kNN/epsilon graph	Explicit graph- geodesic estimator	Weighted graph metric; continuous geodesic estimator under assumptions	Yes, grap geodesic
Directed kNN	Local directed topology	Directed graph distance	Only if asymmet reachabil intended
Symmetrized kNN / iKNN	Local topology before shortest paths	Graph construction	Yes after weights a shortest p
Mutual kNN	Conservative topology	Graph construction	Yes after repair/we if connec
Shared-neighbor graph	Robust clustering topology	Similarity graph	Only afte explicit p cost defir

Family	Primary role	Metric type	Should g call i geodes
Radius graph	Classical random geometric graph	Graph construction	Yes after weights
Hybrid kNN/radius	Topology control	Graph construction	Yes after weights
Local scaling	Multi-scale affinity	Kernel/spectral graph	No, unless converted path cost
Adaptive neighborhoods / IAN	Data-adaptive topology	Graph construction	Yes after weights
MST/CMST	Connectivity scaffold and diagnostic	Connected graph metric	Yes, graph geodesic scaffold
Component repair	Connectivity repair	Graph modification	Yes after weights
Alternative-path pruning	Shortcut/redundancy control	Graph modification	Yes after weights
RNG/Gabriel/beta skeleton	Empty-region geometric pruning	Proximity graph	Yes after weights, cautiously
Delaunay graph	Proximity supergraph	Proximity graph	Yes after weights, cautiously
Geometric spanner	Sparse stretch control	Graph metric preserving chosen base metric	Only if base metric is appropriate
Witness/landmark graph	Scalability/topology surrogate	Sparse graph/complex	Yes after weights, approximate
PAGA/topology abstraction	Topology summary	Cluster/partition graph	Usually not for sample distances
Diffusion distance	Random-walk comparator	Operator/embedding distance	No
Commute/resistance	Electrical/random-walk comparator	Resistance metric / commute time	No
Heat-kernel distance	Multiscale operator geometry	Heat/operator distance	Only small time limit
PHATE potential distance	Biological transition comparator	Diffusion/potential/embedding distance	No
UMAP fuzzy graph/embedding	Fuzzy topology and visualization	Fuzzy graph; embedding distance	No for embedding graph only after path cost defined

Family	Primary role	Metric type	Should g call i geodes
Pseudotime graph distance	Rooted trajectory distance	Graph/curve distance on inferred trajectory	Yes on trajectory graph onl

What Is Known, Borrowed, and Distinctive for gflow

Known and borrowed:

- kNN/epsilon shortest paths are the classical finite-data geodesic estimator.
- asymptotic consistency needs shrinking local neighborhoods, sufficient sampling, and controlled smooth geometry.
- proximity graphs and spanners provide pruning language but not automatic manifold truth.
- single-cell analysis already accepts graph distances, MSTs, principal graphs, and diffusion comparators.

Unsettled:

- which graph topology best serves finite, noisy, uneven biological data;
- when forced connectedness is a feature versus a false bridge;
- how to choose among kNN, mKNN, adaptive graphs, pruning, and repair without overfitting synthetic cases;
- how to define truth when a gap may be biological absence or sampling absence.

Potentially distinctive for gflow:

- treating graph topology, edge weighting, component repair, and path attribution as separate inspectable modules;
- evaluating distances by downstream regression and gradient-flow stability, not only embedding appearance;
- compositional-data-aware neighbor construction for microbiome/metagenomics;
- explicit bridge and shortcut diagnostics on noisy circles, tubes, branches, gaps, and high-dimensional nuisance benchmarks;
- allowing ordinary kNN/mKNN/iKNN to beat CMST when the evidence says they do.

Recommendations for the Benchmark Program

1. Keep the first benchmark axis topology-only: compare kNN, mKNN, radius, hybrid, adaptive, pruned, component-repaired, MST, and CMST graphs with the same Euclidean or compositional edge weights.
2. Separate topology diagnostics from edge-weight diagnostics.
3. Require parameter sweeps, not single-method exemplars.
4. For connectedness repair, report bridge edge identity and shortest-path usage.
5. For pruning, report stretch and false-shortcut effects, not only edge count.
6. Include diffusion, PHATE, UMAP, resistance, and pseudotime as comparators under their correct metric names.

7. In applied microbiome work, benchmark against Aitchison/UniFrac/Bray-Curtis baselines before claiming biological usefulness.
8. In single-cell-style branches, compare all-pairs graph distances to root-based pseudotime and principal-graph distances.
9. For every downstream gflow regression, retain path attribution so distance failures can be traced to specific graph edges.

Source Index

- Classical graph geodesics and Isomap: [Tenenbaum et al. 2000 Science](#), [Bernstein et al. graph approximation report](#), [scikit-learn Isomap docs](#), [scikit-learn manifold guide](#).
- Landmark and out-of-sample Isomap: [de Silva and Tenenbaum Landmark MDS](#), [Bengio et al. NeurIPS 2003](#).
- Local scaling and adaptive neighborhoods: [Self-tuning spectral clustering](#), [IAN Neural Computation](#), [IAN arXiv](#).
- Random geometric graph and kNN graph theory: [Penrose Random Geometric Graphs](#), [randomized near-neighbor graph review](#), [Alamgir and von Luxburg ICML 2012 PDF](#), [Arias-Castro and Le Gouic 2017](#).
- Proximity graphs and pruning: [Relative neighborhood graph overview](#), [Toussaint 1980 record](#), [Gabriel graph overview](#), [beta skeleton overview](#), [Delaunay/RNG/beta skeleton construction](#), [geometric spanner overview](#).
- Single-cell graph adoption: [Monocle 2 / DDRTree](#), [scTEP trajectory review/background](#), [Slingshot methods table](#), [PAGA Genome Biology](#), [VIA background](#), [DensityPath](#).
- Microbiome and bioinformatics adoption: [microarray Isomap example](#), [metagenomic 2D representation example](#), [Scanpy neighbors docs](#), [Seurat FindNeighbors docs](#).
- Spatial/shape examples: [spatial transcriptomics point-cloud graph geodesics](#), [discrete geodesic graph on meshes](#).
- Distinct comparator distances: [Coifman and Lafon diffusion maps](#), [von Luxburg et al. commute/hitting times](#), [PHATE Nature Biotechnology](#), [PHATE docs](#), [UMAP docs](#), [UMAP API fuzzy simplicial set](#).